STORM-PhysNet: A Multi-Horizon Transformer for Geostationary Relativistic Electron Flux Forecasting and Cross-Satellite Transfer

Samarth B N
Dept. of Electronics and Communication Engineering
RV College of Engineering
Bangalore, India
samarthbn.ec25@rvce.edu.in

Samrudh S Malali
Dept. of Electronics and Communication Engineering
RV College of Engineering
Bangalore, India
samrudhsmalali.ec25@rvce.edu.in

Dhyan M
Dept. of Electronics and Communication Engineering
RV College of Engineering
Bangalore, India
dhyanm.ec25@rvce.edu.in

Sanjana H V
Dept. of Artificial Intelligence and Machine Learning Engineering
RV College of Engineering
Bangalore, India
sanjanahv.ci25@rvce.edu.in

***Abstract*—Electrons above 2 MeV at geostationary orbit (GEO) are a recurring internal-charging hazard for communication, navigation and Earth-observation satellites. We present STORM-PhysNet, a Transformer that forecasts GOES-15 >2 MeV log-flux 1, 6 and 12 h ahead from a 72 h history of flux and OMNI solar-wind/geomagnetic drivers, with a learnable propagation delay, a B_z-conditioned gate and residual-over-persistence heads. We evaluate it under a leakage-free protocol (no back-filled gaps, causal Kp, boundary-buffered windows) on three disjoint test periods (2012–13, 2015–16, 2018–20; 34 554 windows, 937 storm-time windows, Dst down to -198 nT), with ten seeds per model and paired tests against an architecture-matched Transformer, a residual-head Transformer, a flux-only Transformer and an LSTM. STORM-PhysNet reaches a block-averaged prediction efficiency of 0.977/0.865/0.803 at 1/6/12 h (persistence 0.968/0.581/0.261). It beats the matched Transformer at 1 h in every block ($+0.008$, $p = 3.5 \times 10^{-18}$), ties it at 6 h and trails it at 12 h (-0.010). Ablations show that the 1 h gain comes from the residual-over-persistence head and the 6–12 h skill from the solar-wind inputs; the delay and gate do not improve clean-data accuracy. They do, however, coincide with much greater robustness: with unit-variance noise on the drivers STORM-PhysNet keeps 1 h PE at 0.970 (matched Transformer 0.892), and on GSAT-19 GRASP data from an Indian-longitude GEO sensor it transfers zero-shot with 12 h PE 0.738 versus 0.644 ($p = 5.3 \times 10^{-4}$). After fine-tuning both models converge (0.802 vs 0.794).**

***Index Terms*—Geostationary orbit, relativistic electrons, space weather, Transformer, data leakage, transfer learning, prediction efficiency, GRASP**

I. Introduction

Relativistic electron fluxes ($E > 2\,\mathrm{MeV}$) at GEO can rise by several orders of magnitude within a day of a geomagnetic disturbance and cause deep dielectric charging of spacecraft electronics [1], [2]. Operators of geostationary communication, navigation and Earth-observation platforms therefore need forecasts at lead times of hours, and increasingly at longitudes, such as the Indian sector, that GOES does not monitor.

Forecasting has progressed from linear prediction filters [3] and empirical models [4] to neural networks [5], [6] and, recently, Transformers [7]–[9]. Two issues limit how these results can be compared and trusted. First, evaluation details matter as much as architecture: interpolated or back-filled gaps, centred filters, interleaved splits and single test years can all inflate skill [10], [11], yet few studies report seed variability or significance tests, and simple baselines often rival Transformers [7], [12]. Second, models are almost always trained and tested on one satellite, while new sensors such as GRASP on GSAT-19 start with short records [13].

This paper makes four contributions. (i) A leakage-free evaluation protocol for hourly GEO flux forecasting with three disjoint test periods spanning the ascending, declining and minimum phases of solar cycle 24, ten seeds per model and paired statistics. (ii) STORM-PhysNet, a multi-horizon Transformer with a learnable delay, a B_z-conditioned gate and residual-over-persistence heads, together with controls that isolate each design choice. (iii) An attribution of skill: residual heads explain the 1 h gain, solar-wind drivers the 6–12 h skill, while the delay and gate do not raise clean-data accuracy but coincide with markedly better robustness to corrupted drivers. (iv) A GOES-15$\rightarrow$GRASP transfer study comparing zero-shot use, heads-only and full fine-tuning, and training from scratch.

II. Related Work

Empirical and neural models. Linear prediction filters relate GEO electron flux to solar-wind speed [3] and underlie NOAA's operational REFM; comparing such models, Perry et al. [4] found no clear winner and a strong solar-cycle dependence of skill. Wei et al. [5] forecast daily >2 MeV flux

one day ahead with an LSTM (PE 0.83–0.91 per test year against 0.68 for persistence). Son et al. [6] forecast hourly flux 1–72 h ahead with a multilayer perceptron on 72 h of solar-wind and index inputs (PE 0.97 at 1 h, 0.53 at 72 h), using a split that interleaves months of the same year. Forsyth et al. [14] framed GOES-15 forecasting as threshold exceedance and found persistence hard to beat at the lower threshold.

Transformers. Tan et al. [7] found a Transformer only slightly, and not significantly, better than TPA-LSTM on GOES-15 flux. Sun et al. [8] compared LSTM and Transformer models on 5-min GOES-10 $\geq$2 MeV flux at 1–24 h (test year 2005; persistence PE 0.913/0.572/0.207 and best Transformer 0.940/0.828/0.747 at 1/6/12 h), with gaps linearly interpolated and inputs selected on that year, and later improved daily fluence prediction with pre-training and few-shot learning [9]. None of these studies reports multi-seed significance tests. In general time-series forecasting, simple linear models can outperform Transformers [12], and the Temporal Fusion Transformer [15] already combines gating with multi-horizon outputs; our gate differs only in being conditioned on B_z.

Evaluation and transfer. Skill should be reported against persistence as well as climatology [10], [16]. Data leakage is a common source of irreproducible ML results [11], and autocorrelated series need temporally blocked validation [17]. Across satellites, Zhao et al. [13] cross-calibrated FY-4A against GOES-15 and trained a nowcasting model on the merged record, which outperformed one trained on GOES-15 alone. We are not aware of a study of fine-tuning regimes for GEO flux forecasting on a new sensor such as GSAT-19 GRASP.

III. Data and Evaluation Protocol

GOES-15 and OMNI. The target is the GOES-15 EPEAD >2 MeV integral electron flux from March 2010 to March 2020, averaged to hourly means (87 168 hours) and $\log_{10}$-transformed. Each forecast origin uses 72 h of history of 18 channels: log-flux, its within-hour standard deviation, an observed-flux mask, and 15 OMNI drivers and derived features (V_{sw}, B_z, B_t, density, P_{dyn}, duration of southward B_z, hours since storm onset, 1/6/24 h means of V_{sw}, 1/6 h means of B_z, B_zV_{sw}, Kp and Dst). OMNI is already propagated from L1 to the bow-shock nose [18].

Leakage controls. An audit of our earlier pipeline found four leakage paths, each removed here. (1) Missing GOES hours (NOAA quality flags or non-positive flux; 27–32% of training hours) were back-filled from later observations; now they are never filled: input histories are forward-filled (past values only), the mask channel marks real hours, and a window is scored only if its last input hour and all three target hours are observed. (2) A centred Hampel despiking filter was removed. (3) OMNI repeats each 3-hourly Kp value on all three hours, exposing up to 2 h of future; each hour now sees only the last completed Kp interval. (4) Each row is labelled train/validation/test and a window is kept only if all 84 rows (72 input plus 12 lead) share one label, giving an 84 h buffer at every boundary. Scalers are fitted on training rows only.

TABLE I
Test blocks. Storm windows have Dst ≤ -50 nT at the 6 h target.

Block	Test period	Train	Test	Gap-free	Storm	Min Dst
A	09/2018–03/2020	38 909	7 536	4 283	18	−69
B	01/2015–12/2016	32 296	14 021	8 452	565	−198
C	01/2012–12/2013	33 413	12 997	7 189	354	−139

Window counts; validation has 9 264 windows in all blocks. Gap-free: no unobserved hour in the 72 h input. Min Dst is over the 6 h targets.

Blocked test periods. The validation period (March 2017 to September 2018) is shared; three test blocks are evaluated separately, each model being retrained with all remaining hours as training data (Table I). Blocks B and C contain the strong storms of 2015 and 2012–13. Because training data then lie on both sides of the test block, this is blocked cross-validation rather than strict forward chaining [17]. Fig. 2 shows the design.

Metrics and statistics. We report prediction efficiency against climatology and persistence [16],

$$\mathrm{PE}_{\mathrm{clim}} = 1 - \mathrm{MSE}(\hat{y}, y)/\mathrm{Var}(y), \tag{1}$$

$$\mathrm{PE}_{\mathrm{pers}} = 1 - \mathrm{MSE}(\hat{y}, y)/\mathrm{MSE}(y_t, y), \tag{2}$$

where y_t is the last observed log-flux. PE without a subscript means $\mathrm{PE}_{\mathrm{clim}}$. Every model is trained with seeds 42–51 in each block (210 GOES runs). Tables give seed means averaged over blocks; comparisons use paired t-tests over the 30 seed–block pairs with Cohen's d_z, and bagged (seed-averaged) predictions get 95% intervals from a 7-day block bootstrap. P-values are not corrected for multiple comparisons.

GRASP. The Geostationary Radiation Spectrometer on GSAT-19 ($\approx$ 82°E; data from ISSDC) provides >2 MeV electron flux from July 2017 to August 2018 (10 249 hours, 10 158 observed; an hour needs at least 6 of its 12 five-minute samples). GRASP is not cross-calibrated to GOES, and we hold all driver channels at their GOES training mean, so the experiment tests transfer of the flux-history model under sensor and longitude shift. Three chronological folds use training from July 2017 up to the fold, one month of validation and a 2–3-month test: G1 February–March 2018 (1 316 test windows, no storm), G2 April–May 2018 (1 381; 13 storm targets) and G3 June–August 2018 (2 109; 47 storm targets from the 25–26 August storm).

IV. Model and Baselines

STORM-PhysNet (Fig. 1; 343 599 parameters) has three additions to a standard Transformer encoder [19] ($d_{\mathrm{model}} = 128$, 2 layers, 4 heads, $d_{\mathrm{ff}} = 256$). *Adaptive delay:* a small network predicts a per-window shift $\tau \in [0.5, 1.5]$ h applied to the driver channels by differentiable interpolation, intended to absorb residual solar-wind timing error. B_z *gate:* an MLP of southward-B_z statistics produces a clamped sigmoid vector $\mathbf{g} \in [g_{\min}, 1]^{128}$ that scales the final encoder state. *Residual*

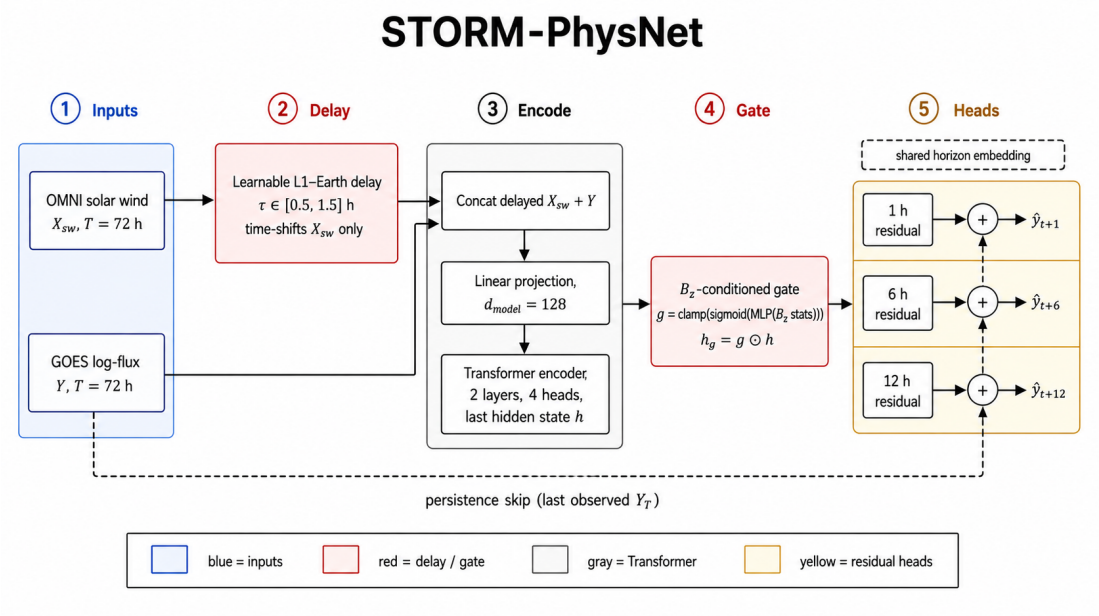


Fig. 1. STORM-PhysNet. OMNI drivers are time-shifted by a learnable delay $\tau \in [0.5, 1.5]$ h, concatenated with the GOES log-flux history ($T = 72$ h), projected to $d_{\text{model}} = 128$ and encoded by a 2-layer, 4-head Transformer. A B_z-conditioned gate scales the final representation and three residual heads add corrections to persistence at 1, 6 and 12 h.

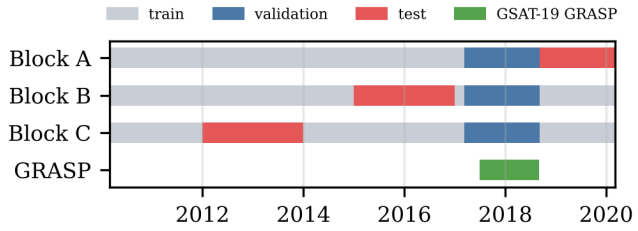


Fig. 2. Evaluation design. Each GOES-15 block is a separate retraining with its own test period and a shared validation period; GRASP (GSAT-19) data cover July 2017 to August 2018.

TABLE II
GOES-15 TEST SKILL, SEED MEANS AVERAGED OVER BLOCKS A–C.

	PE_{clim}, single model			PE_{pers}	PE_{clim}, bagged		
Model	1 h	6 h	12 h	1 h	1 h	6 h	12 h
Persistence	0.968	0.581	0.261	0.00	—	—	—
LSTM	0.937	0.838	0.792	−0.97	0.946	0.860	0.815
TF matched	0.969	0.867	0.813	0.04	0.975	0.884	0.833
TF + residual	0.976	**0.872**	**0.815**	0.25	0.979	**0.888**	**0.836**
TF flux-only	0.968	0.834	0.753	0.01	0.975	0.855	0.779
STORM-PhysNet	0.977	0.865	0.803	0.29	0.979	0.881	0.825
no delay	**0.978**	0.869	0.806	**0.32**	**0.980**	0.883	0.827
no gate	0.977	0.867	0.805	0.28	0.978	0.882	0.828

Seed standard deviations are ≤ 0.008 for STORM-PhysNet and the matched and residual Transformers. Bagged: the ten seed predictions are averaged before scoring. Bold: best per column.

heads: three heads with a shared horizon embedding predict corrections to persistence, $\hat{y}_{t+h} = y_t + \Delta_h$. Training minimises a β-NLL loss [20] on the three horizons with a $2.5\times$ penalty on storm-time under-prediction, plus small auxiliary and regularisation terms (Dst/Kp, smoothness, monotonicity, delay and coupling terms), using AdamW (learning rate 3×10^{-4}), cosine decay, batch 64, storm windows oversampled $12\times$, up to 40 epochs and early stopping on the 12 h validation MSE.

Controls. All controls use the same data, sampler and trainer. *Matched Transformer* (1.19 M): STORM's d_{model}, depth and heads with PyTorch's default $d_{\text{ff}} = 2048$, no delay, gate or residual. *Transformer + residual head* (268 k): $d_{\text{ff}} = 256$ and the persistence residual, but no delay or gate. *Flux-only Transformer*: the matched Transformer with all driver channels set to zero (flux history, its within-hour spread and the mask kept). *STORM ablations*: no delay; no gate. *LSTM* (55 k). Compute was 23 T4 GPU-hours for the GOES runs.

V. Results

A. Skill on GOES-15

Table II and Fig. 3 summarise test skill. All learned models beat persistence clearly at 6 and 12 h, where persistence retains PE of only 0.581 and 0.261. At 1 h persistence is already 0.968, and only models with a persistence residual improve on it substantially ($\text{PE}_{\text{pers}} = 0.25$–$0.32$); the matched Transformer is level with persistence (0.04) and the LSTM is worse (-0.97). The same pattern holds in each block (Fig. 3) and on gap-free windows (STORM-PhysNet 0.980/0.857/0.798 in block A, for example).

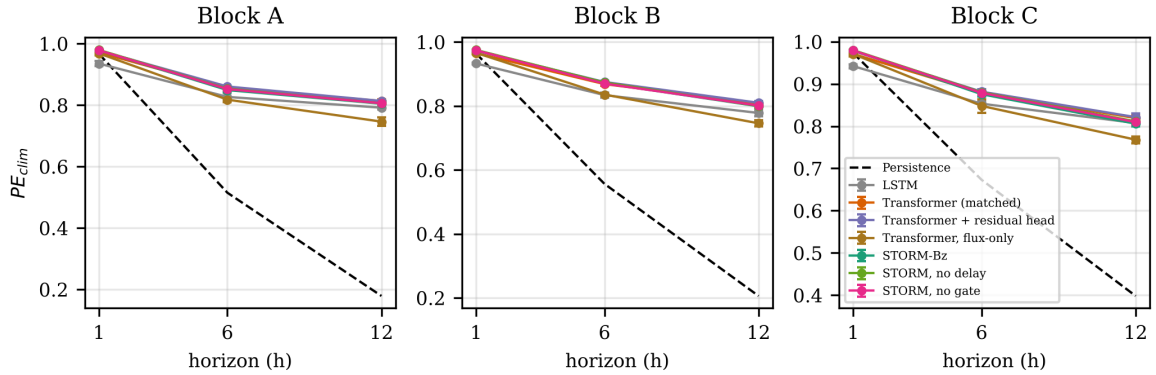


Fig. 3. Test $\mathrm{PE}_{\mathrm{clim}}$ against horizon in each block (mean $\pm$ standard deviation over ten seeds). Dashed: persistence. The learned models are separated mainly from persistence and, at 6–12 h, from the flux-only Transformer.

TABLE III
PAIRED DIFFERENCES IN $\mathrm{PE}_{\mathrm{clim}}$ (30 SEED–BLOCK PAIRS); p-VALUES IN PARENTHESES.

Comparison	1 h	6 h	12 h
STORM − TF matched	+0.0081 (4e-18)	−0.0022 (0.14)	−0.0104 (1e-8)
STORM − TF + residual	+0.0012 (5e-6)	−0.0071 (9e-7)	−0.0118 (4e-8)
TF + residual − TF matched	+0.0069 (4e-17)	+0.0049 (1e-3)	+0.0014 (0.39)
TF matched − flux-only	+0.0008 (0.45)	+0.0336 (5e-17)	+0.0602 (1e-20)
STORM − no delay	−0.0010 (3e-8)	−0.0037 (4e-5)	−0.0026 (0.02)
STORM − no gate	+0.0002 (0.11)	−0.0017 (0.18)	−0.0024 (0.05)
STORM − LSTM	+0.0406 (1e-25)	+0.0273 (9e-12)	+0.0107 (3e-4)

TABLE IV
RMSE OF LOG-FLUX POOLED OVER BLOCKS A–C (SEED MEANS).

	Storm (937 windows)			Non-storm (33 617)		
Model	1 h	6 h	12 h	1 h	6 h	12 h
Persistence	0.301	0.773	0.984	0.142	0.514	0.701
LSTM	0.363	0.597	0.668	0.183	0.290	0.345
TF matched	0.318	0.553	0.613	0.124	0.263	0.329
TF + residual	0.292	**0.548**	**0.606**	0.113	**0.258**	**0.327**
STORM-PhysNet	**0.281**	0.555	0.650	0.114	0.267	0.336
no delay	**0.279***	0.554	0.642	**0.111**	0.264	0.335

Storm: Dst ≤ -50 nT at the 6 h target. Bold: lowest error per column (*ablation, not the proposed model).

B. What Drives the Skill

Table III gives paired differences. *Against the matched Transformer*, STORM-PhysNet is better at 1 h in every block (+0.0067, +0.0091, +0.0085 in A, B, C), statistically tied at 6 h and worse at 12 h in every block (-0.009 to -0.014). The bootstrap intervals of the bagged models agree: the 1 h gain excludes zero in all blocks, the 12 h deficit in B and C.

Residual heads explain the 1 h gain. Adding only a persistence residual to a plain Transformer raises 1 h PE by 0.0069 ($p = 3.8 \times 10^{-17}$) and gives most of STORM-PhysNet's advantage; this smaller control is also the best model at 6 and 12 h. *Solar-wind drivers explain the 6–12 h skill:* removing them leaves 1 h PE unchanged ($p = 0.45$) but costs 0.034 and 0.060 at 6 and 12 h. *The delay and gate do not improve clean-data accuracy:* removing the delay slightly increases PE at all horizons, and removing the gate changes nothing significantly. Fig. 4 shows that these effects have the same sign in every block.

C. Storm Time and Events

Table IV splits the error into the 937 storm-time windows and the rest. During storms STORM-PhysNet (with its no-delay ablation) has the lowest 1 h error, while at 6–12 h all Transformer variants cut the persistence error by 28–38% and the residual Transformer is lowest; at 12 h STORM-PhysNet trails both Transformer controls. For exceedance of $10^3\,\mathrm{cm^{-2}s^{-1}sr^{-1}}$, its 1 h Heidke skill score is 0.925–0.951 across blocks against 0.899–0.928 for persistence, and at 12 h it is 0.757–0.776 against 0.774–0.792 for the matched Transformer. Fig. 5 shows the June 2015 storm: after the dropout, the bagged STORM-PhysNet follows the 24 June enhancement more closely than the matched Transformer, but both miss the peak; this single event is illustrative only.

D. Robustness to Corrupted Drivers

We add Gaussian noise of standard deviation σ (in standardised units) to all driver channels of the test inputs, leaving the flux history unchanged (Fig. 6). At 1 h STORM-PhysNet keeps PE at 0.970 for $\sigma = 1$ and 0.960 for $\sigma = 2$, against 0.949 and 0.895 for the residual Transformer and 0.892 and 0.651 for the matched Transformer. At 6 h and $\sigma = 2$ the values are 0.637, 0.554 and 0.481. Since the residual Transformer degrades faster despite the same persistence anchor, this robustness is the one measurable benefit associated with the delay and gate, possibly because the gate limits how strongly the drivers can perturb the forecast; we did not run the ablations under noise, so the two modules cannot be separated here.

E. Interpretability

The learned delay saturates: its mean is 1.50 h and 92% of test windows (89–97% per block) have $\tau \geq 1.49$ h, so the module acts as a constant extra shift at the upper bound rather than a recovered transit time; because OMNI is already propagated to the bow shock [18], the preference for older

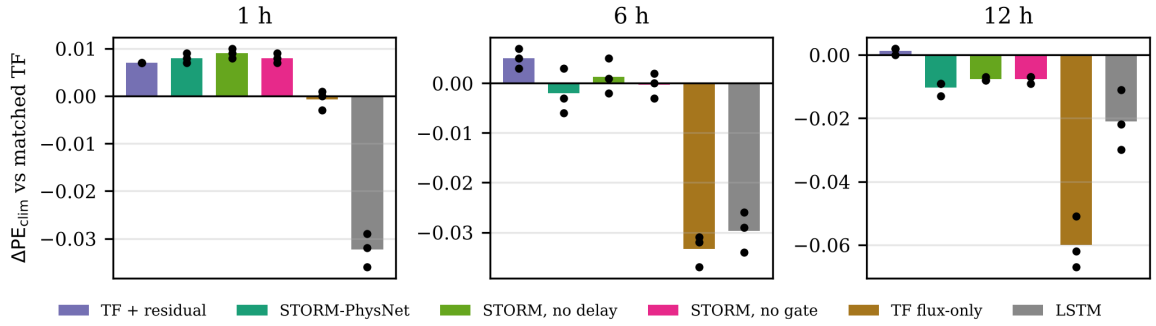


Fig. 4. Difference in $\mathrm{PE}_{\mathrm{clim}}$ from the matched Transformer (bars: mean over blocks; dots: blocks A, B, C; each a ten-seed mean). The persistence residual drives the 1 h gain, the solar-wind inputs the 6–12 h skill; the delay and gate add little.

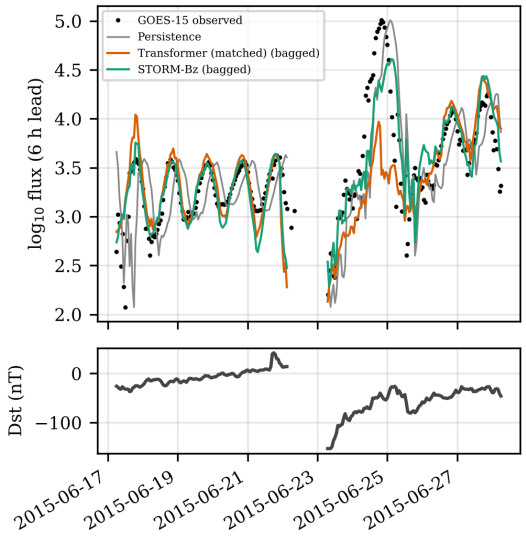


Fig. 5. 6 h forecasts (bagged over ten seeds) during the June 2015 storm, block B. Gaps are unobserved hours; lower panel: Dst.

driver values is plausibly related to the hours-long acceleration timescale [2]. Gate activations are higher in storm than in quiet windows (means 0.441 vs 0.424; area under the ROC curve 0.76 ± 0.10 over the 30 runs), so the gate responds to storm conditions, but the shift is small (Fig. 7).

F. Transfer to GSAT-19 GRASP

Table V and Fig. 8 compare four regimes, starting from the block-A GOES models (persistence on GRASP: 0.951/0.529/0.129). Applied zero-shot, STORM-PhysNet loses little at 1 h and stays far above the matched Transformer at all horizons ($+0.075$, $+0.075$, $+0.095$; $p = 6 \times 10^{-8}$, 2×10^{-5}, 5×10^{-4}); the matched Transformer falls below persistence at 1 h. Fine-tuning the heads alone recovers most of the gap (STORM-PhysNet 12 h PE $+0.049$, $p = 8 \times 10^{-6}$); this comparison is not like-for-like because STORM's heads-only stage trains 63 357 weights against 387 for the Transformer. With full fine-tuning both reach the same skill (differences ≤ 0.008, $p \geq 0.09$), and training on GRASP alone is as good

TABLE V
GRASP TEST $\mathrm{PE}_{\mathrm{clim}}$, MEAN OVER FOLDS G1–G3 (TEN SEEDS).

	STORM-PhysNet			TF matched		
Regime	1 h	6 h	12 h	1 h	6 h	12 h
Zero-shot	0.961	0.827	0.738	0.887	0.751	0.644
Heads-only FT	0.963	0.850	0.787	0.954	0.843	0.773
Full FT	0.964	0.861	0.802	0.964	0.857	0.794
GRASP-only	0.963	0.860	0.799	0.961	0.854	0.790

FT: fine-tuning (Adam, 10^{-4}, horizon-weighted MSE, early stopping). GRASP-only: random initialisation, 3×10^{-4}.

as fine-tuning: with 6–10 months of target data, GOES pre-training brings no further benefit. The value of the GOES model is therefore in the zero- and low-data regime, where the persistence anchor keeps it reliable.

VI. Discussion and Limitations

Our absolute skill is in the range of recent hourly studies (e.g. 0.940/0.828/0.747 at 1/6/12 h in [8]), but cadence, satellite, energy channel and test period differ, so these numbers are not a ranking. Within our protocol the main lesson is that the simplest effective idea, a residual over persistence, is what makes a Transformer useful at short lead, and that physics-motivated modules should be judged by robustness as well as by clean accuracy. Our earlier, leakier pipeline reported 12 h PE 0.854 on the same 2018–20 period as block A, against 0.804 here, which illustrates how strongly evaluation choices affect reported skill.

Operational reading. For a GEO operator the results suggest a simple recipe: anchor every horizon on persistence, rely on solar-wind drivers for 6–12 h warnings, and prefer the gated model where driver data are noisy or missing, or when a new sensor has little history, since it degrades gracefully in both cases. When several months of target data exist, fine-tuning or even target-only training is sufficient.

Limitations. (1) The input Kp and Dst are definitive values; real-time products differ, and solar-wind drivers are not forecast, so 12 h forecasts assume upstream data only up to the forecast time. (2) Blocks B and C train on data from both sides of the test period. (3) Only 937 test windows are storm

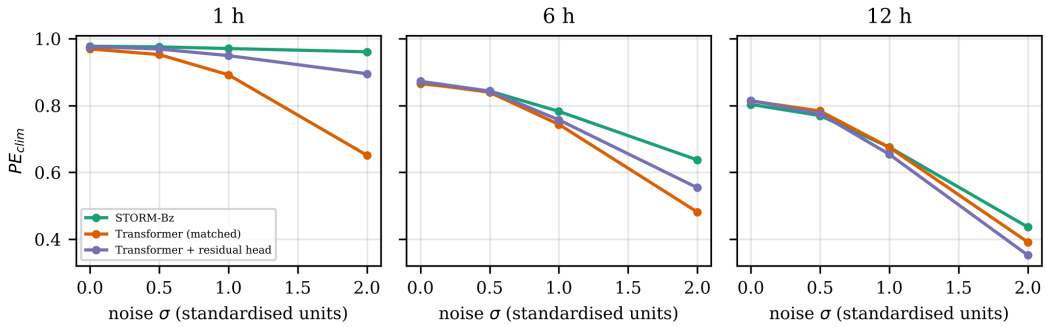


Fig. 6. $\mathrm{PE}_{\mathrm{clim}}$ under Gaussian noise added to the standardised driver channels (seed means averaged over blocks).

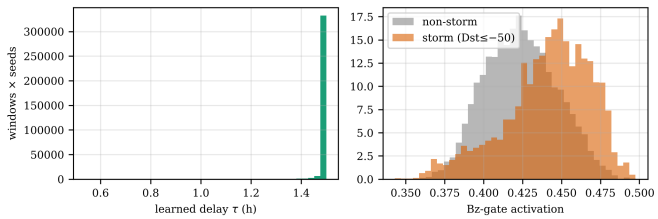


Fig. 7. Left: learned delay over all test windows and seeds, piled at the 1.5 h bound. Right: gate activation in storm and quiet windows (normalised histograms).

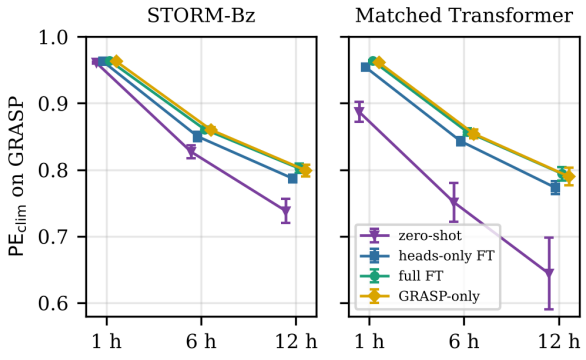


Fig. 8. GRASP skill by regime (mean ± standard deviation over seeds, averaged over folds). Zero-shot (purple) shows the largest difference between models.

time and the case study is a single event. (4) GRASP covers 14 months near solar minimum, is uncalibrated against GOES and was run without driver inputs; the residual-head control was not run on GRASP, so its zero-shot advantage cannot be attributed to the delay and gate rather than the residual. (5) Tests are uncorrected for multiplicity and seeds measure initialisation variance on fixed splits.

VII. Conclusion

Under a leakage-free, multi-period, multi-seed protocol, STORM-PhysNet gives the best 1 h forecasts of GEO >2 MeV flux in every test block and skill far above persistence at 6–12 h, although a simpler residual Transformer is slightly better at those horizons. Its delay and gate do not add clean-data accuracy but coincide with large gains in robustness to corrupted drivers and in zero-shot transfer to the GSAT-19 GRASP sensor, where fine-tuning and target-only training then reach the same skill. We release the protocol, code and all predictions so that future models for Indian-longitude GEO assets can be compared on equal terms.

Data Availability

Code, configurations, the leakage-free data pipeline, per-run predictions, tables and figures are available at https://github.com/bnsama29-cloud/STORM-PhysNet-L1. GOES-15 data are from NOAA NCEI (https://www.ngdc.noaa.gov/stp/satellite/goes/), OMNI from NASA OMNIWeb (https://omniweb.gsfc.nasa.gov/) and GSAT-19 GRASP from ISSDC (https://pradan1.issdc.gov.in/grasp/).

Acknowledgment

The authors thank the GOES, OMNI and GRASP data providers.

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